

PRETRAINING TRANSFORMERS WITH QUANTIZED SOFTMAX IN ATTENTION

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ABSTRACT

Low-precision Transformer systems increasingly quantize attention matrix multiplications, while softmax often remains at higher precision. During pretraining, an approximate softmax changes the gradients that train the model as well as its forward computation. We study this interaction with K -interval attention, which approximates the exponential using $K+1$ grid values. We vary per-row grid calibration, interpolation versus hard rounding, and the placement of a straight-through surrogate relative to normalization. We derive the corresponding backward rules, including calibration derivatives, and compare these choices in pretraining experiments matched on model, data, and optimizer. Detaching the row extrema leaves the forward computation unchanged but produces a delayed increase in validation loss. With hard rounding at $K=4$, min-max calibration and a pre-normalization surrogate incur a large loss gap; changing either choice substantially reduces it. At 124M parameters and 2.5B training tokens, fixed-window calibration with a post-normalization surrogate yields a validation loss gap of +0.019 nats relative to softmax at $K=4$, and with a pre-normalization surrogate yields +0.004 nats at $K=16$.

1 INTRODUCTION

Low-precision Transformer training has moved the linear projections and, increasingly, the attention matrix products to narrow formats, while the exponential and its row reduction are kept in FP16/FP32 by design (Xiao et al., 2023; Mishra et al., 2025; Hernández-Cano et al., 2025; Zhang et al., 2026; Ding et al., 2026). Whether a low-precision reconstruction of the exponential can be used *during pretraining* is left open by several works (Shkolnik et al., 2024; Zhang et al., 2025; Mishra et al., 2025; Ye et al., 2026).

The question differs in kind from the inference-time one. At inference an approximate operator perturbs a fixed function; during training it defines the learning rule, because the model sees the approximate forward *and* is updated by whatever gradient the implementation attaches to it. For softmax, detaching the row maximum is harmless: shift invariance makes its gradient cancel exactly, so subtracting the maximum is treated as a numerical convenience rather than part of the function. The same habit carried over to a calibrated grid is not harmless, because there the row extrema also set the grid; quantization-aware training frameworks, which simulate quantization in floating point (“fake quantization”), likewise keep their observer statistics off the autograd tape (PyTorch contributors, 2026).

We study a family simple enough that every forward and backward term is closed-form (Figure 1, Table 1). Three design axes define an operator: *calibration*, row-wise min-max calibration (MinMax) over each row’s own extrema or a fixed window of τ nats anchored at the row maximum with zero weight below it (FWM); *reconstruction* of the exponential on K intervals from $K+1$ tabulated grid values, by piecewise-linear interpolation (LERP) or by rounding each score to the nearest grid center (Nearest); and *surrogate placement*, Weight-STE on the unnormalized weights before normalization or Prob-STE on the normalized probabilities. We call the family K -interval attention. The experiments study the training properties of coarse reconstruction in fp32 attention arithmetic (TF32 matmuls in training, §4); we measure no kernels and claim no speed-up.